BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: 1481 / 1486

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : (Objective Type)

سوالیہ پرچہ: (معروضی طرز)

QQ-1. 0922 (جماعت نہم) — سیکنڈری پارٹ I — Session 2018-20 to 2021-23 — کیمسٹری (معروضی) — گروپ پہلا

Q. Time: 15 minutes — Total Marks: 12 — PAPER CODE 1481

Note:- You have four choices for each objective type question as A, B, C and D. The choice, which you think, is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of ink remover or white correcting fluid is not allowed

Q. Q-1

Q1. The manufacturing of sulphuric acid is application of

(A) (B)

(C) (D)

Q2. Which one is the example of mixture?

(A) (B)

(C) (D)

Q3. Number of neutrons in deuterium $^{2}_{1}H$ isotope is.

(A) (B)

(C) (D)

Q4. Which one element has the lowest atomic radius?

(A) (B)

(C) (D)

Q5. Polar covalent compounds easily dissolve in

(A) (B)

(C) (D)

Q6. Which of the following has a coordinate covalent bond?

(A) (B)

(C) (D)

Q7. Which one liquid evaporates rapidly?

(A) (B)

(C) (D)

Q8. Transition temperature of tin (Sn) is

(A) (B)

(C) (D)

Q9. Solubility of which one salt increases with the increase in temperature?

(A) (B)

(C) (D)

Q10. Which is an example of reducing agent?

(A) (B)

(C) (D)

Q11. Which one is a non electrolyte?

(A) (B)

(C) (D)

Q12. Metals form ionic compounds with

(A) (B)

(C) (D)

Paper : (Essay Type)

پرچہ: (انشائیہ طرز)

QQ-2. 0922 (**جماعت نہم**) — **سیکنڈری پارٹ** I — Session 2018-20 to 2021-23 — **کیمسٹری (انشائیہ) — حصہ اول — گروپ پہلا**

Q. (Marks 5×2=10) Time: 1:45 hours — Total Marks: 48 — Part I

سوال نمبر 2- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) Define relative atomic mass of an element? Write its unit? (ii) What is meant by mole? Give an example? (iii) How molecular formula is derived from empirical formula? (iv) Write down the electronic configuration of chloride (Cl^-) ion in subshells? (v) Write down two observations made by Rutherford? (vi) Give the trend of electronegativity in a period and a group? (vii) What is meant by atomic radius? Write its unit? (viii) How periods differ from groups?

Q. (Marks 5×2=10) Q-3

سوال نمبر 3- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) Write down any two properties of ionic compounds. (ii) Define hydrogen bonding? (iii) What type of bond is formed in nitrogen molecule? (iv) What is meant by efflusion of gas? (v) State Charle,s law. (vi) Define Transition Temperature with an example. (vii) What do you mean by 22 carat gold? (viii) Write down chemical reaction of Mg with O_2 and N_2

Q. (Marks 5×2=10) Q-4

سوال نمبر 4- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) What is unsaturated solution? How it change into saturated solution? (ii) What is % volume/mass? (iii) Why the suspension does not form homogeneous mixture? (iv) Why ether, benzene and petrol are insoluble in water? (v) Define redox reaction. (vi) What is Alloying? (vii) Whose are non electrolytes? (viii) What is the basic function of salt bridge?

Q. (Marks 9 × 2 = 18) Part II — Note: Attempt any Two Questions.

نوٹ: کوئی سے دو سوالات کے جوابات تحریر کریں۔

Q5. 5.(a) Write any three results of the experiments and two defects of Rutherford Atomic Model? (3+2)

(b) Define Molecular Ion? Write any three differences of Molecule and Molecular Ion? (1+3)

Q6. 6.(a) What is a coordinate covalent bond? Explain it with an example (2+3)

(b) What is Boyle's law and also derive its equation. (2+2)

Q7. 7.(a) Explain the construction and working of Daniel Cell, draw diagram also. (2+2+1)

(b) How does Sodium react with Water, Oxygen, Hydrogen and Chlorine? Write equations. (1x4)

Note: Board typos kept verbatim: 'Charle,s law', 'homogencous'.

GROUP – II

Q. Paper : (Objective Type)

سوالیہ پرچہ: (معروضی طرز)

QQ-1. 0922 (جماعت نہم) — سیکنڈری پارٹ I — Session 2018-20 to 2021-23 — کیمسٹری (معروضی) — گروپ دوسرا

Q. Time: 15 minutes — Total Marks: 12 — PAPER CODE 1486

Note:- You have four choices for each objective type question as A, B, C and D. The choice, which you think, is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of ink remover or white correcting fluid is not allowed

Q. Q-1

Q1. Escaping of gas molecules through a tiny hole in the space with lesser pressure is called

(A) (B)

(C) (D)

Q2. A solution formed by dissolving a substance in water is called.

(A) (B)

(C) (D)

Q3. Slow and continuous eating away of a metal by the surrounding medium is called

(A) (B)

(C) (D)

Q4. The oxidation number of oxygen in peroxides is

(A) (B)

(C) (D)

Q5. The most abundant metal is

(A) (B)

(C) (D)

Q6. The development of chemical industry has generated.

(A) (B)

(C) (D)

Q7. The study of manufacturing of chemical compounds on commercial base is

(A) (B)

(C) (D)

Q8. Almost all the particles passed through the foil undeflected. This observation was made by.

(A) (B)

(C) (D)

Q9. Creater of first version of Periodic table of elements

(A) (B)

(C) (D)

Q10. Chlorine atom has electrons in the valence shell

(A) (B)

(C) (D)

Q11. The chemical bond by mutual sharing of their valence shell electrons is called

(A) (B)

(C) (D)

Q12. The scale of temperature in Kelvin Scale starts from °C?

(A) (B)

(C) (D)

Paper : (Essay Type)

پرچہ: (انشائیہ طرز)

QQ-2. 0922 (**جماعت نہم**) — **سیکنڈری پارٹ** I — Session 2018-20 to 2021-23 — **کیمسٹری (انشائیہ) — حصہ اول — گروپ دوسرا**

Q. (Marks 5×2=10) Time: 1:45 hours — Total Marks: 48 — Part I

سوال نمبر 2- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) Write down the enpirical formula of silica and benzene. (ii) Define triatomic molecule with an example. (iii) Calculate the number of moles in '6' gram of water. (iv) Write two observations of Rutherford's experiment. (v) Define electronic configuration. (vi) State

modern periodic law. Who put farward this law? (vii) What are the trends of sheilding effect in the periodic table. (viii) Why elements of group 1st and 2nd are called S-block elements

Q. (Marks 5×2=10) Q-3

سوال نمبر 3- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) Write two properties of non polar covalent compounds. (ii) What is meant by single covalent bond? Give an example (iii) Show intermolecular and bonding forces in HCl? (iv) Define allotropy? Write any one reason of allotropy. (v) State Boyle's Law? Write its mathematical expression? (vi) What is meant by transition temperature? Give an example? (vii) Give the trend of electropositivity in a group and period? (viii) Write any two physical characteristics of metals?

Q. (Marks 5×2=10) Q-4

سوال نمبر 4- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) How heat is given out in solubility? (ii) Differentiate between colloid and solution. (iii) How one molar solution is prepared? (iv) What is meant by m/v % ? (v) What is corrosion? Give its example. (vi) Write the function of Salt bridge? (vii) Define Alloy with example. (viii) What is stainless steel?

Q. (Marks 9 × 2 = 18) Part II — Note: Attempt any Two Questions.

نوٹ: کوئی سے دو سوالات کے جوابات تحریر کریں۔

Q5. 5(a) Explain five differences between Rutherford Atomic Theory and Bohr's Atomic theory. 1x5=5

(b) Can you identify mixture, Element and Compound out of the following 0.5x8=4 (i) Tin (ii) Coca cola (iii) Ice Crcain (iv) Petroleum (v) Gun powder (vi) Sugar (vii) Salt (viii) Blood

Q6. 6(a) Define ionic compounds and write their three properties 2+3=5

(b) How boiling point of liquid depends on nature of liquid and intermolecular forces. 2+2=4

Q7. 7(a) What is the principle of Electroplating? How electroplating of chromium is carried out? 2+3=5

(b) Write any four chemical properties of Non-metals? 1x4=4

> Note: Board typos kept verbatim: 'enpirical', 'farward', 'sheilding', 'Ice Crcain'.